

2024 年全国硕士研究生招生考试数学一解析

一、选择题

1. 答 应选 C.

解 $e^{\cos t}$ 关于 t 是偶函数, 则 $\int_0^x e^{\cos t} dt$ 是奇函数. 由 $g(x) = \int_0^{\sin x} e^{t^2} dt$, 则 $g(-x) = \int_0^{\sin(-x)} e^{t^2} dt = \int_0^{-\sin x} e^{t^2} dt$, 令 $t = -u$, 则 $g(-x) = -\int_0^{\sin x} e^{u^2} du$, 于是 $g(-x) = -g(x)$, $g(x)$ 是奇函数.

注 $g(x) = \int_0^{\sin x} e^{t^2} dt$ 可以看作由 $y(u) = \int_0^u e^{t^2} dt$ 和 $u(x) = \sin x$ 复合而成, 显然, $u(x) = \sin x$ 是奇函数, 于是 $g(x)$ 的奇偶性与 $y(u)$ 一致, 是奇函数. 另外, 若 $u(x)$ 是偶函数, 则 $g(x)$ 直接就是偶函数 (此时与 $y(u)$ 的奇偶性无关).

2. 答 应选 A.

解 由转换投影公式, 知

$$\begin{aligned} \iint_{\Sigma} P dydz + Q dzdx &= \iint_{\Sigma} P \cdot \left(-\frac{\partial z}{\partial x}\right) dx dy + Q \cdot \left(-\frac{\partial z}{\partial y}\right) dx dy \\ &= \iint_{\Sigma} \left(P \cdot \frac{x}{z} + Q \cdot \frac{y}{z}\right) dx dy = \iint_{\Sigma} \left(\frac{x}{z} P + \frac{y}{z} Q\right) dx dy. \end{aligned}$$

3. 答 应选 A.

解 $\ln(2+x) = \ln\left(1+\frac{x}{2}\right) + \ln 2 = \ln 2 + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\left(\frac{x}{2}\right)^n}{n} = \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \cdot 2^n} x^n, -2 < x \leq 2$.

依题意, 有 $a_0 = \ln 2, a_n = \frac{(-1)^{n-1}}{n \cdot 2^n}, n = 1, 2, \dots$, 进而 $a_{2n} = \frac{(-1)^{2n-1}}{2n \cdot 2^{2n}} = -\frac{1}{2n \cdot 4^n}, n = 1, 2, \dots$, 则

$$\sum_{n=0}^{\infty} n a_{2n} = \sum_{n=1}^{\infty} n a_{2n} = -\sum_{n=1}^{\infty} n \cdot \frac{1}{2n \cdot 4^n} = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{4^n} = -\frac{1}{2} \cdot \frac{\frac{1}{4}}{1 - \frac{1}{4}} = -\frac{1}{6}.$$

4. 答 应选 B.

解 由 $f'(0) = m$, 则 $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = m$, 即 $\lim_{x \rightarrow 0} [f(x) - f(0)] = 0$, 又 $\lim_{x \rightarrow 0} f(x) = 0$, 于是 $f(0) = 0$.

此时 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = m$.

5. 答 应选 B.

解 由题意可知, π_1, π_2, π_3 相交于一条直线, 且不重合, 于是方程组
$$\begin{cases} a_1 x + b_1 y + c_1 z = d_1, \\ a_2 x + b_2 y + c_2 z = d_2, \\ a_3 x + b_3 y + c_3 z = d_3 \end{cases}$$
 有无穷多

解, 且 $\alpha_1, \alpha_2, \alpha_3$ 两两不相关, 故 $r \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = r \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} < 3$, 且 $r(\alpha_i, \alpha_j) = 2 (i \neq j)$, 故 $m = n = 2$.

6. 答 应选 D.

$$\text{解 } (\alpha_1, \alpha_2, \alpha_3) = \begin{pmatrix} a & 1 & 1 \\ 1 & 1 & a \\ -1 & b & -1 \\ 1 & a & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1-a & 1-a^2 \\ 0 & b+1 & a-1 \\ 0 & a-1 & 1-a \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1-a & 1-a^2 \\ 0 & b+1 & a-1 \\ 0 & 0 & 2-a^2-a \end{pmatrix}.$$

因为 $\alpha_1, \alpha_2, \alpha_3$ 线性相关, 且其中任意两个向量均线性无关, 则 $r(\alpha_1, \alpha_2, \alpha_3) \leq 2$, 且 $r(\alpha_i, \alpha_j) = 2 (i \neq j)$, 于是 $r(\alpha_1, \alpha_2, \alpha_3) = 2$.

① 当 $a = 1$ 时, α_1 与 α_3 线性相关, 不满足题意;

② 当 $a \neq 1$ 时,

$$(\alpha_1, \alpha_2, \alpha_3) \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1 & 1+a \\ 0 & b+1 & a-1 \\ 0 & 0 & a+2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & a \\ 0 & 1 & 1+a \\ 0 & 0 & -b(a+1)-2 \\ 0 & 0 & a+2 \end{pmatrix},$$

要满足题意, 则 $a+2=0$ 且 $-b(a+1)-2=0$, 故 $\begin{cases} a=-2, \\ b=2. \end{cases}$

7. 答 应选 A.

解 由 $r(A) = 2$, 则 $Ax = 0$ 只有一个线性无关的解, 故特征值 0 只有一个线性无关特征向量.

由 $A\alpha = 0 = 0 \cdot \alpha, \alpha \neq 0$, 知 α 为 $\lambda_1 = 0$ 的特征向量.

对 3 维非零列向量, 且满足 $\beta^T \alpha = 0$ (即 α 与 β 正交) 的线性无关向量 β 应当有两个 (比如 3 维坐标系的三个坐标轴), 设为 β_1, β_2 , 由 $A\beta_i = \beta_i = 1 \cdot \beta_i, i = 1, 2$, 则 β_1, β_2 为特征值 $\lambda_2 = \lambda_3 = 1$ 的线性无关特征向量, 进而 A^n 的全部特征值也是 0, 1, 1. 所以 $\text{tr}(A^n) = 0 + 1 + 1 = 2$.

8. 答 应选 B.

解 $E(2X+Y) = 2EX + EY = -2, D(2X+Y) = 4DX + DY = 4 \times 2 + 2 = 10$, 则 $2X+Y \sim N(-2, 10)$. 同理, $X-Y \sim N(2, 4)$. 此时

$$P\{2X+Y < a\} = P\left\{\frac{2X+Y+2}{\sqrt{10}} < \frac{a+2}{\sqrt{10}}\right\} = \Phi\left(\frac{a+2}{\sqrt{10}}\right),$$

且 $P\{X > Y\} = P\left\{\frac{X-Y-2}{2} > \frac{0-2}{2}\right\} = 1 - \Phi(-1) = \Phi(1)$, 由 $\frac{a+2}{\sqrt{10}} = 1$, 得 $a = \sqrt{10} - 2$.

9. 答 应选 D.

$$\text{解 由题意可知 } f(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{其他}, \end{cases} \quad f_{Y|X}(y|x) = \begin{cases} \frac{1}{1-x}, & x < y < 1, \\ 0, & \text{其他}. \end{cases}$$

$$\text{于是 } f(x, y) = f(x) \cdot f_{Y|X}(y|x) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{其他}. \end{cases}$$

$$EX = \int_0^1 2x(1-x)dx = 2 \times \left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3},$$

$$EY = \int_0^1 dx \int_x^1 2ydy = \int_0^1 (1-x^2)dx = \frac{2}{3},$$

$$E(XY) = \int_0^1 dx \int_x^1 2xy dy = \int_0^1 x(1-x^2) dx = \frac{1}{4},$$

$$\text{于是 } \text{Cov}(X, Y) = E(XY) - EX \cdot EY = \frac{1}{4} - \frac{2}{9} = \frac{1}{36}.$$

10. 答 应选 D.

$$\text{解 } X \text{ 与 } Y \text{ 的联合概率密度为 } f(x, y) = f_X(x) \cdot f_Y(y) = \begin{cases} \lambda^2 e^{-\lambda(x+y)}, & x > 0, y > 0, \\ 0, & \text{其他.} \end{cases}$$

设 Z 的分布函数为 $F_Z(z)$, 则 $F_Z(z) = P\{Z \leq z\} = P\{|X - Y| \leq z\}$.

① 当 $z < 0$ 时, $F_Z(z) = 0$;

② 当 $z \geq 0$ 时,

$$\begin{aligned} F_Z(z) &= P\{-z \leq X - Y \leq z\} = 2P\{0 \leq X - Y \leq z\} \\ &= 2 \int_0^{+\infty} \lambda e^{-\lambda y} dy \int_y^{y+z} \lambda e^{-\lambda x} dx \\ &= 2 \int_0^{+\infty} \lambda e^{-\lambda y} [e^{-\lambda y} - e^{-\lambda(y+z)}] dy \\ &= 2 \int_0^{+\infty} \lambda e^{-2\lambda y} dy - 2e^{-\lambda z} \int_0^{+\infty} \lambda e^{-2\lambda y} dy \\ &= 1 - e^{-\lambda z}. \end{aligned}$$

所以 $Z \sim E(\lambda)$, 从而 Z 与 X 服从相同的分布.

二、填空题

11. 答 应填 6.

$$\text{解 } \lim_{x \rightarrow 0} \frac{(1+ax^2)^{\sin x} - 1}{x^3} = \lim_{x \rightarrow 0} \frac{e^{\sin x \ln(1+ax^2)} - 1}{x^3} = \lim_{x \rightarrow 0} \frac{\sin x \ln(1+ax^2)}{x^3} = \lim_{x \rightarrow 0} \frac{ax^3}{x^3} = 6,$$

所以 $a = 6$.

12. 答 应填 5.

解 由 $df \Big|_{(1,1)} = 3du + 4dv$, 则 $f'_u(1,1) = 3, f'_v(1,1) = 4$, 由 $y = f(\cos x, 1+x^2)$, 则

$$\frac{dy}{dx} = f'_1 \cdot (-\sin x) + f'_2 \cdot 2x,$$

$$\frac{d^2 y}{dx^2} = [f''_{11} \cdot (-\sin x) + f''_{12} \cdot 2x](-\sin x) + f'_1 \cdot (-\cos x) + [f''_{21} \cdot (-\sin x) + f''_{22} \cdot 2x] \cdot 2x + f'_2 \cdot 2,$$

$$\text{因此 } \frac{d^2 y}{dx^2} \Big|_{x=0} = f'_1(1,1) \cdot (-1) + f'_2(1,1) \cdot 2 = -3 + 8 = 5.$$

13. 答 应填 $-\frac{1}{\pi}$.

解 这是余弦级数, 则

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi (x+1) \cos nx dx = \frac{2}{\pi} \int_0^\pi x \cos nx dx = \frac{2}{n\pi} \int_0^\pi x d(\sin nx) \\ &= \frac{2}{n\pi} \left(x \sin nx \Big|_0^\pi - \int_0^\pi \sin nx dx \right) = \frac{2}{n\pi} \cdot \frac{1}{n} \cos nx \Big|_0^\pi \end{aligned}$$

$$= \frac{2}{n^2\pi} [(-1)^n - 1].$$

于是 $a_{2n-1} = \frac{2}{(2n-1)^2\pi} [(-1)^{2n-1} - 1] = -\frac{4}{(2n-1)^2\pi}$, 此时

$$\lim_{n \rightarrow \infty} n^2 \sin a_{2n-1} = \lim_{n \rightarrow \infty} n^2 \cdot \sin \frac{-4}{(2n-1)^2\pi} = \lim_{n \rightarrow \infty} n^2 \cdot \frac{-4}{(2n-1)^2\pi} = -\frac{1}{\pi}.$$

14. 答 应填 $\arctan(x+y) = y + \frac{\pi}{4}$.

解 令 $u = x+y$, 则 $y = u-x$, $y' = \frac{du}{dx} - 1$. 代入原方程, 得 $\frac{du}{dx} - 1 = \frac{1}{u^2}$. 于是

$$\frac{du}{dx} = \frac{1}{u^2} + 1 = \frac{1+u^2}{u^2},$$

分离变量, 得 $\frac{u^2}{1+u^2} du = dx$, 两边积分, 得 $\int \frac{u^2}{1+u^2} du = \int dx$, 于是 $u - \arctan u = x + C$. 回代, 得 $x+y - \arctan(x+y) = x + C$, 即

$$y - \arctan(x+y) = C,$$

将 $x=1, y=0$ 代入, 得 $C = -\frac{\pi}{4}$, 于是所求特解为 $\arctan(x+y) = y + \frac{\pi}{4}$.

15. 答 应填 $[0, +\infty)$.

解 由题意, $\mathbf{A}$ 是实对称矩阵, 则存在正交矩阵 $\mathbf{Q}$, 使 $\mathbf{Q}^T \mathbf{A} \mathbf{Q} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \mathbf{\Lambda}$, 故 $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$.

对任意 α, β , 有 $(\alpha^T \mathbf{A} \beta)^2 \leq \alpha^T \mathbf{A} \alpha \beta^T \mathbf{A} \beta$, 即 $(\alpha^T \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T \beta)^2 \leq \alpha^T \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T \alpha \beta^T \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T \beta$ 成立.

记 $\mathbf{Q}^T \alpha = \alpha_1 = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$, $\mathbf{Q}^T \beta = \beta_1 = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, 则 $(\alpha_1^T \mathbf{\Lambda} \beta_1)^2 \leq \alpha_1^T \mathbf{\Lambda} \alpha_1 \beta_1^T \mathbf{\Lambda} \beta_1$, 即

$$(\lambda_1 a_1 b_1 + \lambda_2 a_2 b_2)^2 \leq (\lambda_1 a_1^2 + \lambda_2 a_2^2)(\lambda_1 b_1^2 + \lambda_2 b_2^2),$$

也即 $\lambda_1^2 a_1^2 b_1^2 + \lambda_2^2 a_2^2 b_2^2 + 2\lambda_1 \lambda_2 a_1 b_1 a_2 b_2 \leq \lambda_1^2 a_1^2 b_1^2 + \lambda_2^2 a_2^2 b_2^2 + \lambda_1 \lambda_2 a_1^2 b_2^2 + \lambda_1 \lambda_2 a_2^2 b_1^2$,

于是 $2\lambda_1 \lambda_2 a_1 b_1 a_2 b_2 \leq \lambda_1 \lambda_2 a_1^2 b_2^2 + \lambda_1 \lambda_2 a_2^2 b_1^2$, 即 $\lambda_1 \lambda_2 (a_1^2 b_2^2 + a_2^2 b_1^2 - 2a_1 b_1 a_2 b_2) \geq 0$, 也即 $\lambda_1 \lambda_2 (a_1 b_2 - a_2 b_1)^2 \geq 0$,

由 α, β 的任意性, 可知 $\lambda_1 \lambda_2 \geq 0$, 于是 $|\mathbf{A}| = \begin{vmatrix} a+1 & a \\ a & a \end{vmatrix} = a^2 + a - a^2 = a \geq 0$.

16. 答 应填 $\frac{2}{3}$.

解 设事件 A 为 3 次试验全部成功, 事件 B 为至少成功 1 次, 则

$$\begin{aligned} P(A | B) &= \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} \\ &= \frac{p^3}{1 - (1-p)^3} \\ &= \frac{4}{13}, \end{aligned}$$

于是 $13p^3 = 4 - 4(1-p)^3$, 整理得 $p(3p-2)(3p+6) = 0$, 解得 $p = \frac{2}{3}$.

三、解答题

17. 解 记 $D_1 = \{(x, y) \mid \sqrt{1-y^2} \leq x \leq 1, 0 \leq y \leq 1\}$, 则

$$\begin{aligned} \iint_{D_1} \frac{x}{\sqrt{x^2+y^2}} dx dy &= 2 \iint_{D_1} \frac{x}{\sqrt{x^2+y^2}} dx dy \\ &= 2 \int_0^{\frac{\pi}{4}} \cos \theta d\theta \int_1^{\frac{1}{\cos \theta}} r dr + 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos \theta d\theta \int_1^{\frac{1}{\sin \theta}} r dr \\ &= \int_0^{\frac{\pi}{4}} \left(\frac{1}{\cos^2 \theta} - 1 \right) \cos \theta d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(\frac{1}{\sin^2 \theta} - 1 \right) \cos \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} (\sec \theta - \cos \theta) d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(\frac{\cos \theta}{\sin^2 \theta} - \cos \theta \right) d\theta \\ &= - \int_0^{\frac{\pi}{2}} \cos \theta d\theta + \int_0^{\frac{\pi}{4}} \sec \theta d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\sin^2 \theta} d(\sin \theta) \\ &= -1 + \ln(\sqrt{2} + 1) + (-1 + \sqrt{2}) \\ &= \ln(\sqrt{2} + 1) + \sqrt{2} - 2. \end{aligned}$$

18. 解 (1) 因为 $f(x, y) = x^3 + y^3 - (x+y)^2 + 3$, 所以

$$\frac{\partial f}{\partial x} = 3x^2 - 2(x+y), \frac{\partial f}{\partial y} = 3y^2 - 2(x+y), \frac{\partial f(1,1)}{\partial x} = -1, \frac{\partial f(1,1)}{\partial y} = -1.$$

所以 Γ 的方程为 $x-1+y-1+z-1=0$, 即 $x+y+z=3$.

(2) 由(1)知, Γ 与坐标平面所围有界区域在 xOy 平面上的投影

$$D = \{(x, y) \mid 0 \leq x \leq 3, 0 \leq y \leq 3-x\}.$$

$$\text{令} \begin{cases} \frac{\partial f}{\partial x} = 0, \\ \frac{\partial f}{\partial y} = 0, \end{cases} \quad \text{即} \begin{cases} 3x^2 - 2(x+y) = 0, \\ 3y^2 - 2(x+y) = 0, \end{cases} \quad \text{解得} \begin{cases} x = 0, \\ y = 0 \end{cases} \quad \text{或} \begin{cases} x = \frac{4}{3}, \\ y = \frac{4}{3}. \end{cases}$$

当 $(x, y) \in \{(x, y) \mid 0 \leq x \leq 3, y = 0\}$ 时, $f(x, 0) = x^3 - x^2 + 3, x \in [0, 3]$.

令 $\frac{d}{dx}[f(x, 0)] = 3x^2 - 2x = 0$, 得 $x = 0$ 或 $x = \frac{2}{3}$.

当 $(x, y) \in \{(x, y) \mid 0 \leq y \leq 3, x = 0\}$ 时, 同理可得 $y = 0$ 或 $y = \frac{2}{3}$.

当 $(x, y) \in \{(x, y) \mid 0 \leq x \leq 3, x+y=3\}$ 时, $f(x, 3-x) = x^3 + (3-x)^3 - 6, x \in [0, 3]$.

令 $\frac{d}{dx}[f(x, 3-x)] = 3x^2 - 3(3-x)^2 = 0$, 得 $x = \frac{3}{2}$.

综上, $f(x, y)$ 在 D 上的可能最值点是 $(0, 0), (\frac{4}{3}, \frac{4}{3}), (\frac{2}{3}, 0), (0, \frac{2}{3}), (3, 0), (0, 3), (\frac{3}{2}, \frac{3}{2})$.

又因为 $f(0, 0) = 3, f(\frac{4}{3}, \frac{4}{3}) = \frac{17}{27}, f(\frac{2}{3}, 0) = f(0, \frac{2}{3}) = \frac{77}{27}, f(3, 0) = f(0, 3) = 21, f(\frac{3}{2}, \frac{3}{2}) =$

$\frac{3}{4}$, 所以 $f(x, y)$ 在 D 上的最大值为 21, 最小值为 $\frac{17}{27}$.

19. 证 (1) 因为 $f(x)$ 具有 2 阶导数, 当 $x \in (0, 1)$ 时, 根据泰勒公式, 存在 $\xi_1, \xi_2 \in (0, 1)$, 使得

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(\xi_1)x^2,$$

$$f(x) = f(1) + f'(1)(x-1) + \frac{1}{2}f''(\xi_2)(x-1)^2,$$

所以

$$(x-1)f(x) = f(0)(x-1) + f'(0)x(x-1) + \frac{1}{2}f''(\xi_1)x^2(x-1),$$

$$xf(x) = f(1)x + f'(1)(x-1)x + \frac{1}{2}f''(\xi_2)(x-1)^2x.$$

因为 $f'(0) = f'(1)$, 所以

$$f(x) = f(0)(1-x) + f(1)x + \frac{x(1-x)}{2}[xf''(\xi_1) + (1-x)f''(\xi_2)].$$

又因为 $|f''(x)| \leq 1$, 所以当 $x \in (0, 1)$ 时,

$$|f(x) - f(0)(1-x) - f(1)x| \leq \frac{x(1-x)}{2}[x|f''(\xi_1)| + (1-x)|f''(\xi_2)|] \leq \frac{x(1-x)}{2}.$$

(2) 因为 $\int_0^1 [f(x) - f(0)(1-x) - f(1)x] dx = \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2}$, 所以由定积分的性质和(1)

可得

$$\begin{aligned} \left| \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2} \right| &\leq \int_0^1 |f(x) - f(0)(1-x) - f(1)x| dx \\ &\leq \int_0^1 \frac{x(1-x)}{2} dx \\ &= \frac{1}{12}. \end{aligned}$$

20. 解 记曲线 L 在平面 $2x - z - 1 = 0$ 上所围部分为 Σ , 上侧为正. 由斯托克斯公式得

$$\begin{aligned} &\int_L (6xyz - yz^2) dx + 2x^2 z dy + xyz dz \\ &= \iint_{\Sigma} (xz - 2x^2) dy dz + (6xy - 3yz) dz dx + (z^2 - 2xz) dx dy \\ &= \frac{1}{\sqrt{5}} \iint_{\Sigma} [-2(xz - 2x^2) + (z^2 - 2xz)] dS \\ &= \frac{1}{\sqrt{5}} \iint_{\Sigma} (2x - z)^2 dS \\ &= \frac{1}{\sqrt{5}} \iint_{\Sigma} dS \\ &= \frac{4\sqrt{5}}{25} \pi. \end{aligned}$$

21. 解 由题设得 $\begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix} = \begin{pmatrix} -2 & 0 & 2 \\ 0 & -2 & -2 \\ -6 & -3 & 3 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ y_{n-1} \\ z_{n-1} \end{pmatrix}$, 得矩阵 $A = \begin{pmatrix} -2 & 0 & 2 \\ 0 & -2 & -2 \\ -6 & -3 & 3 \end{pmatrix}$ 满足 $\alpha_n = A\alpha_{n-1}$.

因为

$$|\lambda E - A| = \begin{vmatrix} \lambda + 2 & 0 & -2 \\ 0 & \lambda + 2 & 2 \\ 6 & 3 & \lambda - 3 \end{vmatrix} = \lambda(\lambda - 1)(\lambda + 2),$$

所以矩阵 $\mathbf{A}$ 的特征值为 $\lambda_1 = 0, \lambda_2 = 1, \lambda_3 = -2$.

$$\text{当 } \lambda_1 = 0 \text{ 时, 解方程组 } (0\mathbf{E} - \mathbf{A})\mathbf{x} = \mathbf{0}, \text{ 得特征向量 } \boldsymbol{\xi}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix};$$

$$\text{当 } \lambda_2 = 1 \text{ 时, 解方程组 } (\mathbf{E} - \mathbf{A})\mathbf{x} = \mathbf{0}, \text{ 得特征向量 } \boldsymbol{\xi}_2 = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix};$$

$$\text{当 } \lambda_3 = -2 \text{ 时, 解方程组 } (-2\mathbf{E} - \mathbf{A})\mathbf{x} = \mathbf{0}, \text{ 得特征向量 } \boldsymbol{\xi}_3 = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}.$$

$$\text{令 } \mathbf{P} = (\boldsymbol{\xi}_1, \boldsymbol{\xi}_2, \boldsymbol{\xi}_3) = \begin{pmatrix} 1 & 2 & -1 \\ -1 & -2 & 2 \\ 1 & 3 & 0 \end{pmatrix}, \text{ 则 } \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \text{ 即}$$

$$\mathbf{A} = \mathbf{P} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \mathbf{P}^{-1},$$

从而得

$$\begin{aligned} \mathbf{A}^n &= \mathbf{P} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}^n \mathbf{P}^{-1} = \begin{pmatrix} 1 & 2 & -1 \\ -1 & -2 & 2 \\ 1 & 3 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (-2)^n \end{pmatrix} \begin{pmatrix} 6 & 3 & -2 \\ -2 & -1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -4 - (-2)^n & -2 - (-2)^n & 2 \\ 4 - (-2)^{n+1} & 2 - (-2)^{n+1} & -2 \\ -6 & -3 & 3 \end{pmatrix}. \end{aligned}$$

$$\text{由递推式 } \boldsymbol{\alpha}_n = \mathbf{A}\boldsymbol{\alpha}_{n-1} \text{ 知 } \boldsymbol{\alpha}_n = \mathbf{A}^n\boldsymbol{\alpha}_0, \text{ 其中 } \boldsymbol{\alpha}_0 = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}, \text{ 所以}$$

$$\boldsymbol{\alpha}_n = \mathbf{A}^n\boldsymbol{\alpha}_0 = \begin{pmatrix} -4 - (-2)^n & -2 - (-2)^n & 2 \\ 4 - (-2)^{n+1} & 2 - (-2)^{n+1} & -2 \\ -6 & -3 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 + (-2)^n \\ -8 + (-2)^{n+1} \\ 12 \end{pmatrix},$$

故 $x_n = 8 + (-2)^n, y_n = -8 + (-2)^{n+1}, z_n = 12 (n = 1, 2, \dots)$.

$$22. \text{ 解 } (1) X \text{ 的分布函数为 } F(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{\theta}, & 0 \leq x < \theta, \\ 1, & x \geq \theta, \end{cases} \text{ 则 } X_{(n)} \text{ 的概率密度为}$$

$$f_{(n)}(x) = n[F(x)]^{n-1}F'(x) = \begin{cases} \frac{n}{\theta^n}x^{n-1}, & 0 \leq x \leq \theta, \\ 0, & \text{其他,} \end{cases}$$

从而

$$EX_{(n)} = \int_{-\infty}^{+\infty} xf_{(n)}(x)dx = \int_0^\theta \frac{n}{\theta^n} x^n dx = \frac{n}{n+1}\theta.$$

故 $ET_c = \frac{cn}{n+1}\theta$, 所以当 $c = \frac{n+1}{n}$ 时, T_c 是 θ 的无偏估计.

(2) 由(1) 知

$$EX_{(n)}^2 = \int_{-\infty}^{+\infty} x^2 f_{(n)}(x)dx = \int_0^\theta \frac{n}{\theta^n} x^{n+1} dx = \frac{n}{n+2}\theta^2,$$

$$h(c) = E(T_c - \theta)^2 = \theta^2 \left(\frac{n}{n+2}c^2 - \frac{2n}{n+1}c + 1 \right),$$

所以当 $c = \frac{n+2}{n+1}$ 时, $h(c)$ 最小.